Altered rectal sensory response induced by balloon distention in patients with functional abdominal pain syndrome

Background Functional abdominal pain syndrome (FAPS) has chronic unexplained abdominal pain and is similar to the psychiatric diagnosis of somatoform pain disorder. A patient with irritable bowel syndrome (IBS) also has chronic unexplained abdominal pain, and rectal hypersensitivity is observed in a majority of the patients. However, no reports have evaluated the visceral sensory function of FAPS precisely. We aimed to test the hypothesis that FAPS would show altered visceral sensation compared to healthy controls or IBS. The present study determined the rectal perceptual threshold, intensity of sensation using visual analogue scale (VAS), and rectal compliance in response to rectal balloon distention by a barostat in FAPS, IBS, and healthy controls. Methods First, the ramp distention of 40 ml/min was induced and the thresholds of discomfort, pain, and maximum tolerance (mmHg) were measured. Next, three phasic distentions (60-sec duration separated by 30-sec intervals) of 10, 15 and 20 mmHg were randomly loaded. The subjects were asked to mark the VAS in reference to subjective intensity of sensation immediately after each distention. A pressure-volume relationship was determined by plotting corresponding pressures and volumes during ramp distention, and the compliance was calculated over the linear part of the curve by calculating from the slope of the curve using simple regression. Results Rectal thresholds were significantly reduced in IBS but not in FAPS. The VAS ratings of intensity induced by phasic distention (around the discomfort threshold of the controls) were increased in IBS but significantly decreased in FAPS. Rectal compliance was reduced in IBS but not in FAPS. Conclusion An inconsistency of visceral sensitivity between lower and higher pressure distention might be a key feature for understanding the pathogenesis of FAPS.